



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

Fax: +66-2721-3515

SAFETY DATA SHEET

Company Identification

Thep Watana Co., Ltd. Address: 382 Bangpoo Industrial Estate.

Sukhumvit Road, Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66(0)-2721-3510;

Fax: +66(0)-2721-3515

1. CHEMICAL IDENTIFICATION

Product Identification

Product name: carbendazim 50% W/V SC

Product use: Fungicide

Relevant identified uses of the substance/mixture and uses advised against:

Used as systemic fungicide with protective and curative action. Carbendazim works by inhibiting the development of fungi probably by interfering with spindle formation at mitosis (cell division).

It is recommended to control blast and sheath blight of paddy, loose smut of wheat and barley, brown rot of maize, powdery mildew of sugar beet, peas, cluster beans, cucurbits, rose and ber, anthracnose of grapes and tobacco, seeding blight of jute and downy spot of walnut, damping off of chilies and scab of apples etc.

2. HAZARD IDENTIFICATION

Adverse Human Health Effects

Single ingestion,

The absorption of carbendazim can possibly have irreversible effects.

Short-term Eye contact,

Non-irritant.

Short-term Skin contact,

Non-irritant, sensitization dermatitis may occur in persons previously exposed.

Short-term Inhalation,

The absorption of carbendazim can possibly have irreversible effects.

Environmental effects TOXIC to aquatic organisms.



Thep Watana Co., Ltd.

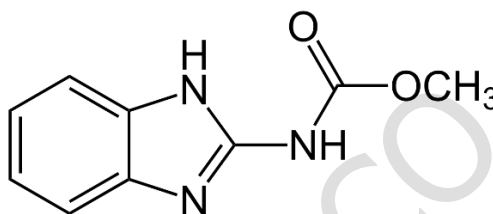
382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

Fax: +66-2721-3515

3. COMPOSITION/INFORMATION ON INGREDIENT

Formulation type:	Suspension Concentrate
Chemical name:	Methyl benzimidazol-2-ylcarbamate (IUPAC) or carbendazim (E-ISO)
Common name:	carbendazim
CAS number:	10605-21-7
EEC number:	234-232-0
Ingredients and composition:	Active ingredient – carbendazim 500 g/L SC
Chemical family:	benzimidazole
Formula:	$C_9H_9N_3O_2$
Molecular weight:	191.2 g/mol



4. FIRST AID MEASURES

Description of first aid measures

In case of swallowed:

Remove by gastric lavage and catharsis. Maintain blood pressure and airway. Give oxygen if respiration is depressed. Do not perform gastric lavage if victim is unconscious. Get medical attention immediately.

In case of eyes contact:

Rinse immediately with plenty of water and seek medical advice.

In case of skin contact:

Remove all contaminated clothes and shoes. Wash off immediately with soap and plenty of water (15 - 20 minutes). Get medical attention if irritation persists.

In case of Inhaled contact:

Remove from exposure area to fresh air immediately. If breathing has stopped, perform artificial respiration. Keep person warm and at rest. Treat symptomatically and supportively. Get medical attention immediately.

Other information

Treat symptomatically and supportively.



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

Fax: +66-2721-3515

5. FIRE FIGHTING MEASURES

Extinguishing Media:

Dry chemical, carbon dioxide, halon, water spray, or standard foam.

Unsuitable extinguishing:

High pressure water jet media.

Special exposure hazards:

Do not release chemically contaminated water into drains, soil and surface waters.

Special protective:

As for all fires involving chemicals: chemical protection suit, suitable gloves equipment for fire-fighters and boots, and self-contained breathing apparatus.

6. ACCIDENTAL RELEASE MEASURES (SPILLAGE)

Personal precautions:

Use personal protective equipment.

Methods for cleaning up:

Do not touch spilled material. Stop leak if you can do it without risk.

Procedure for spill:

For small spills, take up with sand or other absorbent material and place into containers for later disposal. For small dry spills, with a clean shovel place material into clean, dry containers and cover. Move containers from spill area. For larger spills, dike far ahead of spill for later disposal. Keep unnecessary people away. Isolate hazard area and deny entry. Ventilate closed space.

Environmental precautions:

Prevent contamination of water.

7. HANDLING AND STORAGE

Good industrial practice in housekeeping and personal hygiene should be followed.

Handling Precautions:

Keep product in original tightly closed container.

Avoid exposure by inhalation.

Do not get on skin, eyes or clothing.

Do not eat, drink or smoke when handling.

Wash hands before eating, drinking or smoking.

Storage Conditions:

Storage Store in locked, well-ventilated storeroom, away from children and unauthorized personnel, and food and animal feed.



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

Fax: +66-2721-3515

8. EXPOSURE CONTROL/PERSONAL PROTECTION

Exposure control limits: -

Ventilation

Provide local exhaust or general dilution ventilation.

Engineering controls

Respiratory Protection:

Wear a suitable mask.

Hand Protection:

Wear rubber or plastic gloves.

Eye Protection:

Wear splash-proof or dust-resistant safety goggles.

Skin and Body Protection:

Wear overalls and closed footwear.

9. PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Physical stage:	Viscous liquid
Color:	White
Odor:	Light
pH:	6.9 – 7.1 (1% solution)
Boiling point:	100°C (water based formulation)
Melting/ Freezing Point:	Not applicable
Combustibility:	Combustible C1
Flash point:	Not applicable (water based formulation)
Flammability:	Not flammable
Oxidizing properties:	None
Vapor pressure:	Not applicable
Density:	1.175
Solubility in water:	Not applicable
Partition coefficient:	K_{ow} log P = 1.38 (pH 5) ; log P = 1.51 (pH 7) ; log P = 1.45 (pH 9)



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

Fax: +66-2721-3515

10. STABILITY AND REACTIVITY

Stability:

Stable under normal temperatures and pressures.

Conditions to avoid:

No special precautions other than good housekeeping of chemicals.

Material to avoid:

Alkaline materials can cause hydrolysis.

Hazardous decomposition:

Thermal decomposition products may include toxic oxides of carbon and nitrogen.

11. TOXICOLOGICAL INFORMATION

Information on Acute toxicity for end product, Tetraconazole:

LD50 (Oral, rat): > 2,000 mg/kg

LD50 (Dermal, rat): > 2,000 mg/kg

LC50 (Inhalation, rat, 4h): 5.9 mg/L

Skin irritation (rabbit): Not irritant.

Eye irritation (rabbit): Not irritant.

Skin sensitization: Not determined, but carbendazim is not a sensitizer.

Carcinogenicity data

No carcinogenic effect

Mutagenicity data

Carbendazim is not mutagenic but phenazines, possible contaminants of the technical material, are mutagenic at very low concentrations.

Reproductive/Teratology data

Not teratogenic

ADI (man)

0.03 mg/kg b.w.

12. ECOLOGICAL INFORMATION

Toxicity

Adopt good working practices, so that the product is not released into the environment.

Data referred to the mixture, Carbendazim:

Birds – Acute toxicity:

Bobwhite quail,

LD50: 5,826 mg/kg



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

Fax: +66-2721-3515

Fish – Acute toxicity:

Rainbow trout,

LC50 (96 h): 0.83 mg/L

Daphnia magna (a small freshwater crustacean),

LC50 (48 h): 87 ug/L

Honey bee – Acute toxicity:

NOT TOXIC to bee

LC50: 50 ug/bee

Soil – Acute toxicity:

DT50 3 – 12 months (with its residues and metabolites tightly bound to soil)

13. DISPOSAL CONSIDERATIONS

Pesticide disposal:

Bury in a specially designated landfill (soil burial should suffice, do not contaminate water sources). Always dispose of in accordance with local regulations.

Dispose of empty, used containers by:

Packaging should be disposed of at a specialized chemical waste disposal facility.

14. TRANSPORT INFORMATION

The data provided in this section is for information only. Please apply the appropriate regulations to properly classify your shipment for transportation.

UN No: 3082

Dangerous Goods Class: 9

Packaging Group: III

Proper shipping name: Environmentally hazardous substance, liquid, n.o.s. (carbendazim).

Environmental risk: Marine pollutant

15. REGULATORY INFORMATION

Hazard symbol: The harmful symbol (Xn); mutagenic Category 3

The dangerous for the environment symbol (N)

Risk Phrases:

R40/22: Harmful: possible risks of irreversible effects if swallowed.

R50: Very toxic to aquatic organisms.

Safety Phrases:

S13 - Keep away from food, drink or feedingstuffs



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

Fax: +66-2721-3515

S20/21 - When using do not eat, drink or smoke.

S36/37 - Wear suitable protective clothing and gloves.

S46 - If swallowed, seek medical advice immediately and show this container or label.

S60 - This material and/or its containers must be disposed of as hazardous waste.

S61 - Avoid release to the environment. Refer to special instructions/safety data sheet

16. OTHER INFORMATION

MSDS last revision date 06.07.99

The information contained in this safety data sheet is given in good faith. It is accurate to the best of our knowledge and belief and represents the most up to date information. The information given in this data sheet does not constitute or replace the user's own assessment of work place risk as required by other health and safety legislation.